

作业7

1.

驱动: $J_1 = K_1 = 1$

$$J_2 = A \oplus Q_1$$

$$K_2 = A \oplus Q_1$$

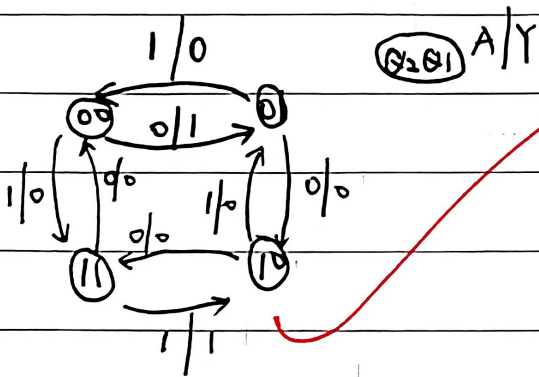
状态方程: $Q_1^* = \bar{Q}_1$

$$Q_2^* = A \oplus Q_1 \oplus Q_2$$

输出方程

$$Y = A Q_1 Q_2 + \bar{A} \bar{Q}_1 \bar{Q}_2$$

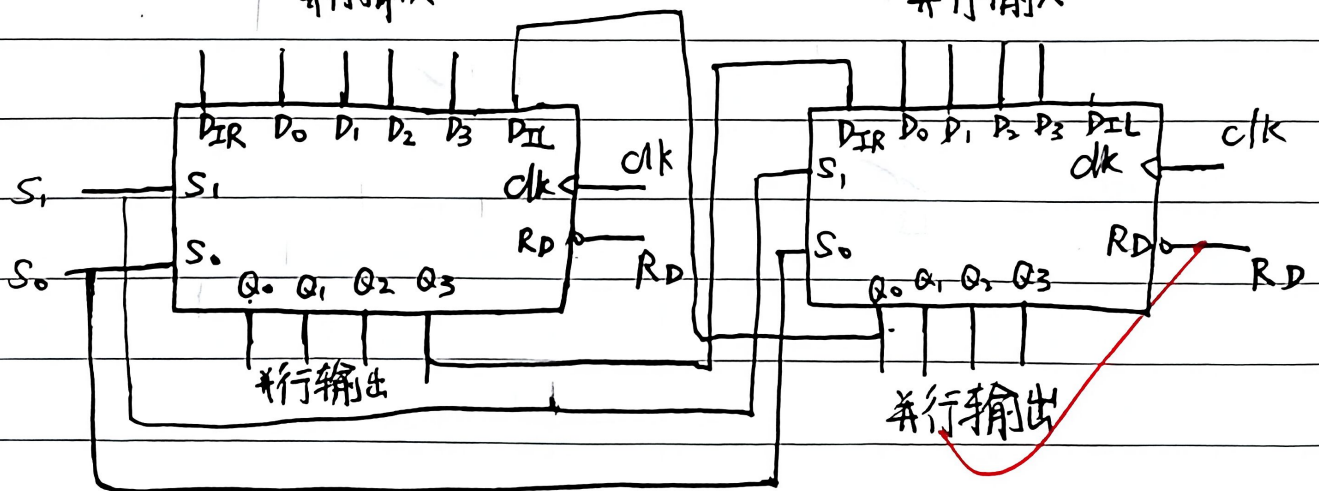
A	Q₁	Q₂	Q₁[*]	Q₂[*]	Y	A / Q₂[*] Q₁[*] / Y	Q ₂ Q ₁	00	01	10	11
1	0	0	1	1	1	1					
1	0	1	1	1	0	0		01 / 1	10 / 0	11 / 0	00 / 0
1	1	0	0	1	0	1		11 / 0	00 / 0	01 / 0	10 / 1
1	1	1	0	0	1	1					



2.

并行输入

并行输入

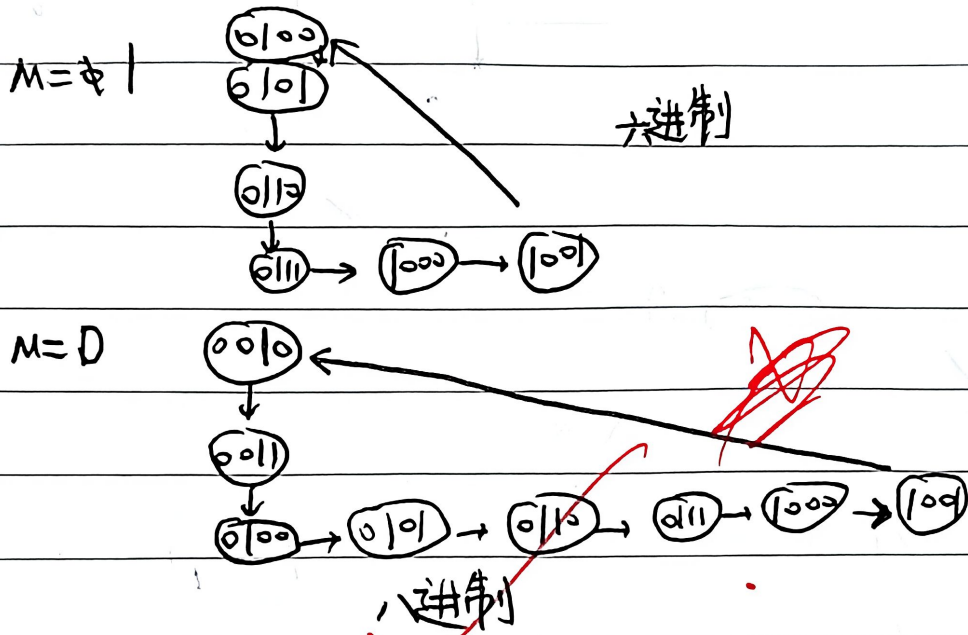


3.

$M=0$ 时预置数 $D_3 D_2 D_1 D_0$ 为 0010

$M=1$ 时为 0100

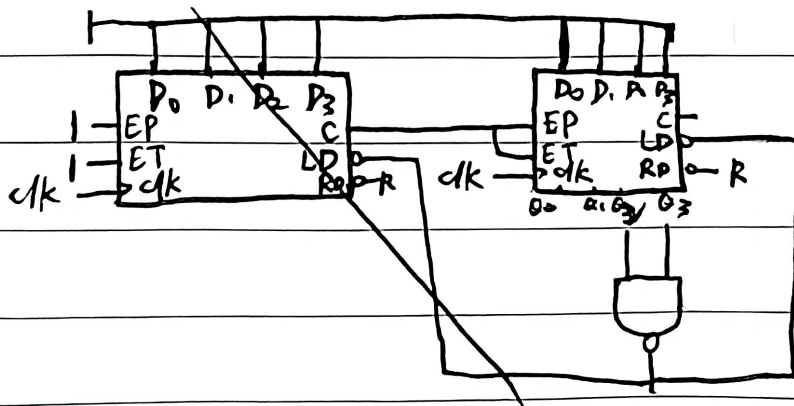
当 $Q_3 Q_2 Q_1 Q_0 = 1 \times 1$ 时 $\overline{LD} = 0$ $LD = 1$

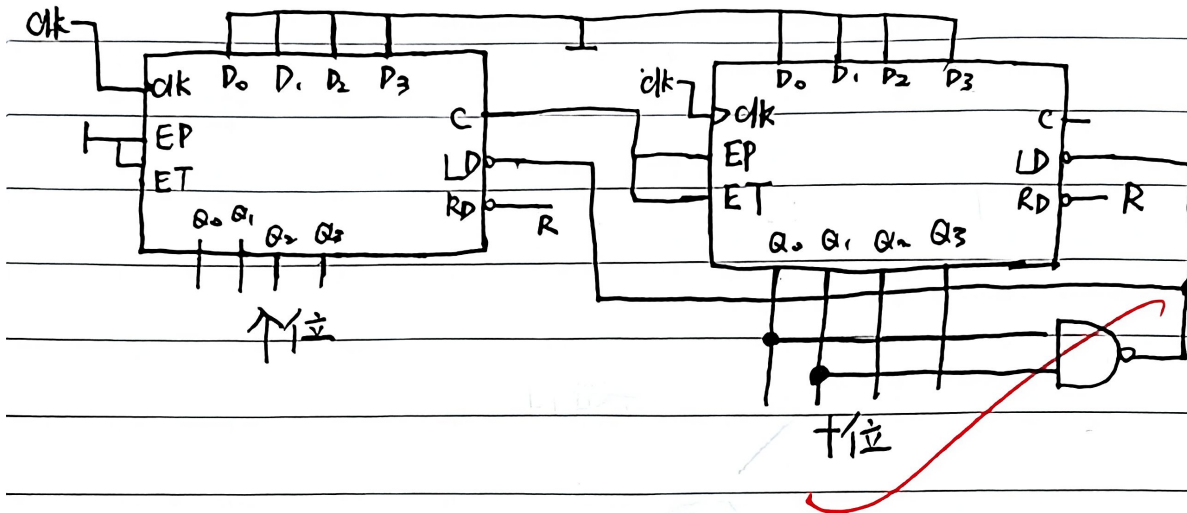


4. 状态定为 0, 1, 2, ..., 30

$D_3 D_2 D_1 D_0 = 0000$

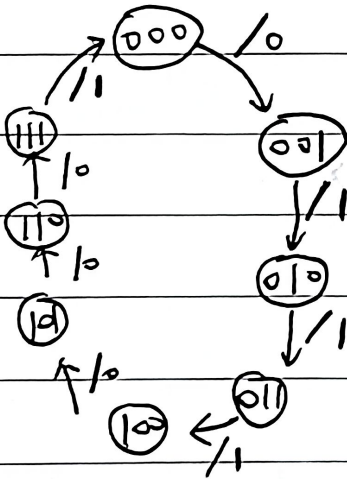
采用置数法, $LD = 1$ 当高位 $Q_3 Q_2 Q_1 Q_0 = 0011$





5.

$Q_2 Q_1 Q_0 / Y$



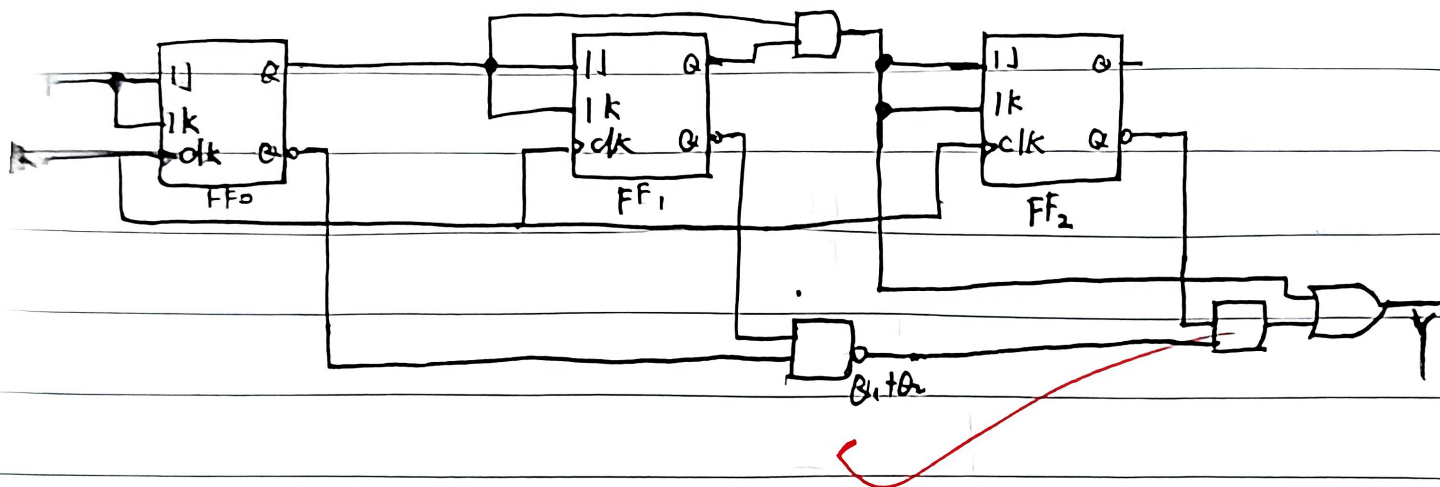
Q_2	$Q_1^* Q_0^*$	$Q_1 Q_0$	00	01	11	10
0			001/0	010/1	100/1	110/1
1			101/0	110/0	000/1	111/0

$$Q_2^* = \bar{Q}_2 Q_1 Q_0 + Q_2 \bar{Q}_1 + Q_2 \bar{Q}_0 \quad J_2 = Q_1 Q_0 \quad K_2 = Q_1 Q_0$$

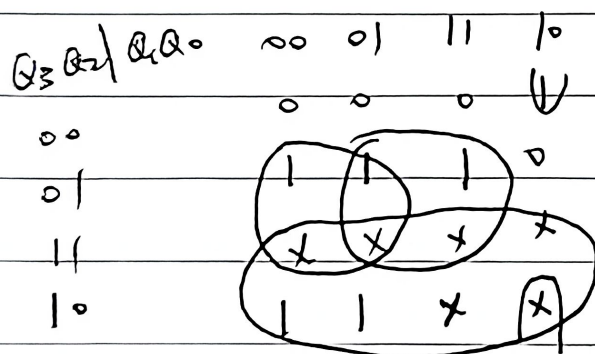
$$Q_1^* = \bar{Q}_1 Q_0 + Q_1 \bar{Q}_0 \quad J_1 = Q_0 \quad K_1 = Q_0$$

$$Q_0^* = \bar{Q}_0 \quad J_0 = K_0 = 1$$

$$Y = \bar{Q}_2 Q_0 + \bar{Q}_2 Q_1 + Q_1 Q_0$$



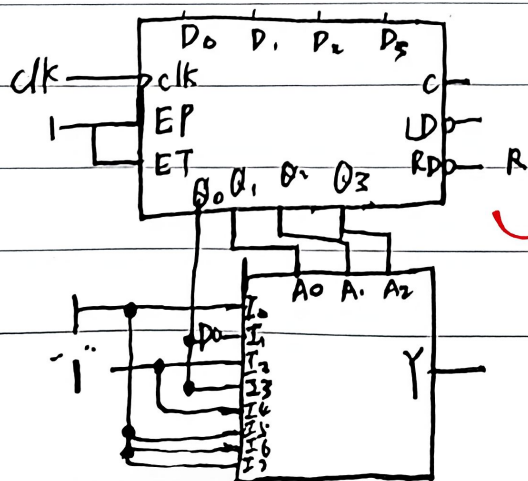
b.



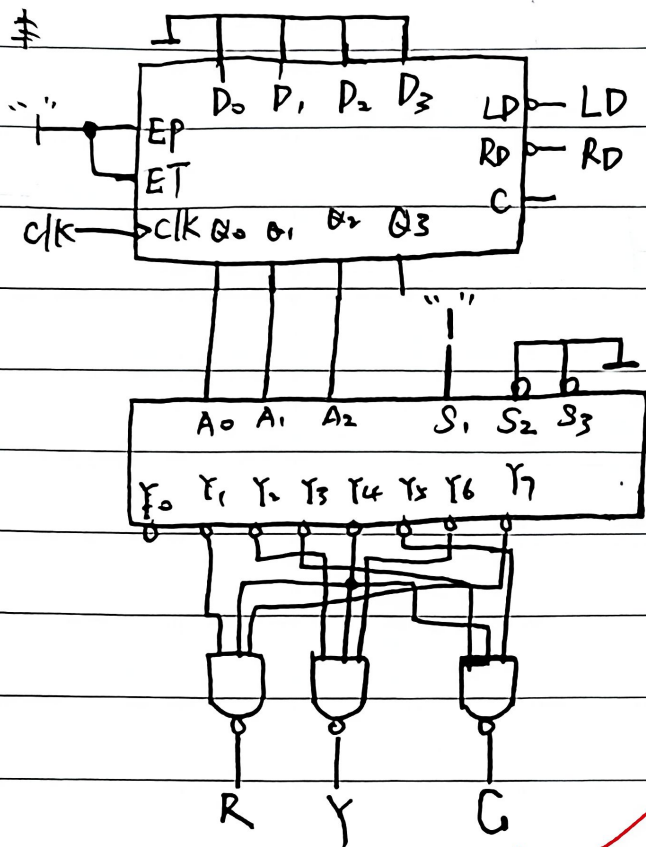
$$Y = Q_3 + Q_2 \bar{Q}_1 + Q_2 Q_0 + \bar{Q}_2 Q_1 \bar{Q}_0$$

令 MUX 的 $A_2, A_1, A_0 = Q_3, Q_2, Q_1$,

$$I_0 = I_5 = I_6 = I_7 = 0 \quad I_2 = I_4 = 1 \quad I_1 = \bar{Q}_0 \quad I_3 = Q_0$$



7. 74LS161 计数器产生 8 个状态, 再经过三线八线译码器产生最小项
经与非门产生控制信号



$$R = m_1 + m_4 + m_7$$

$$Y = m_2 + m_4 + m_6$$

$$C = m_3 + m_4 + m_5$$

8.

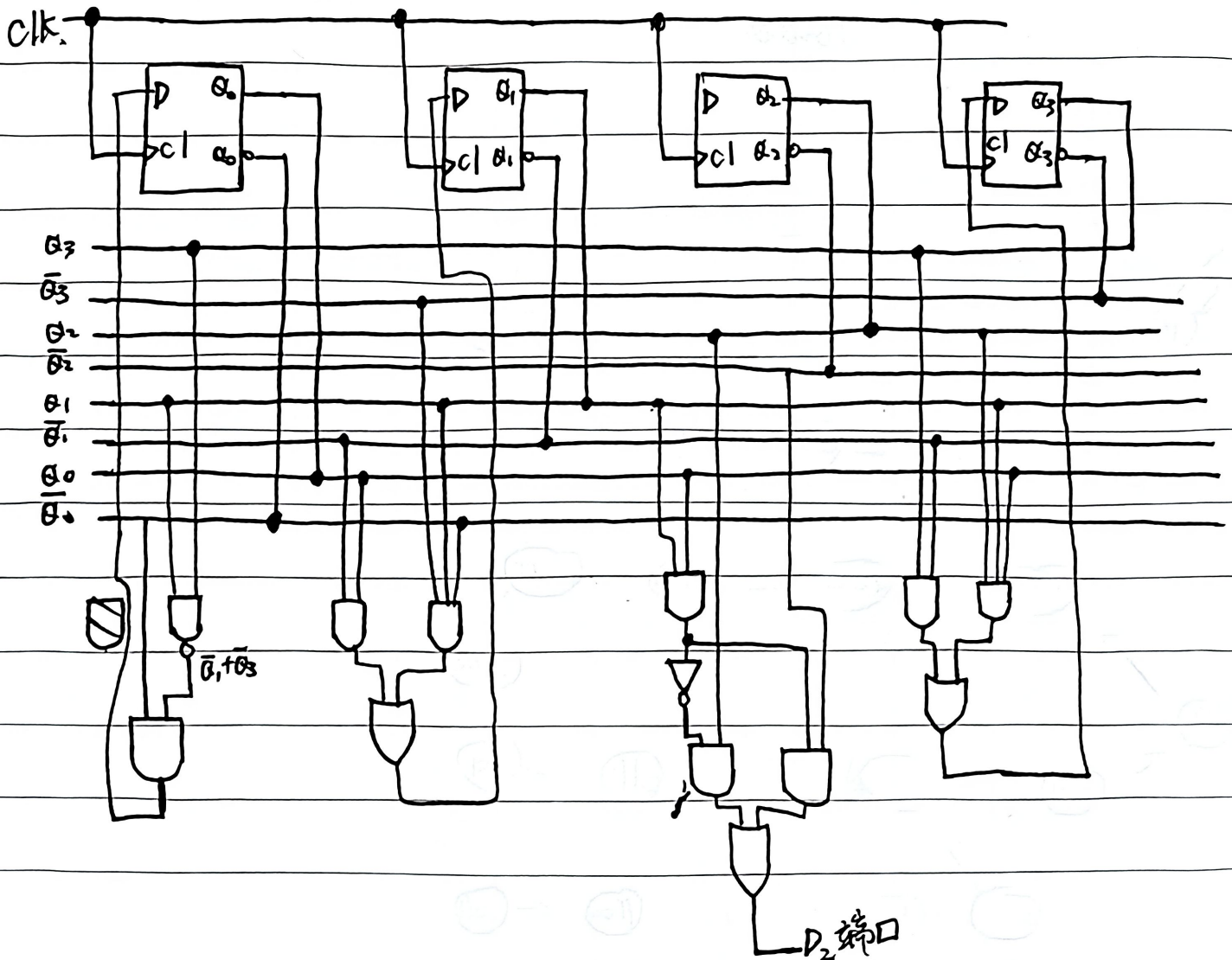
$Q_3 Q_2 / Q_1 Q_0$	00	01	11	10
00	0001	0010	0100	0011
01	0101	0110	1000	0111
11	x	x	x	x
10	1001	1010	x	0000

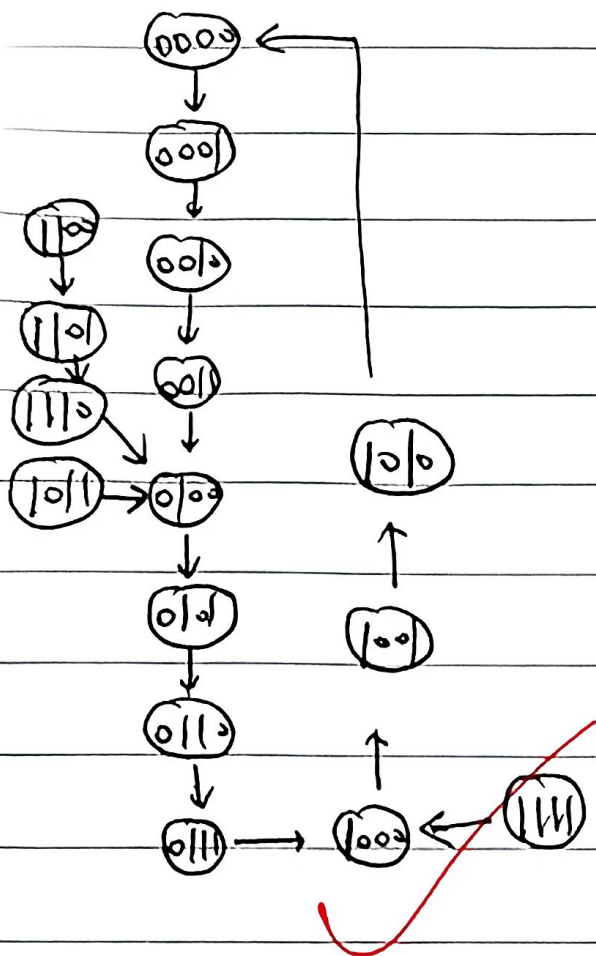
$$Q_0^* = \bar{Q}_3 \bar{Q}_0 + \bar{Q}_1 \bar{Q}_0$$

$$Q_1^* = \bar{Q}_1 Q_0 + \bar{Q}_3 Q_1 \bar{Q}_0$$

$$Q_2^* = \bar{Q}_2 Q_1 Q_0 + Q_2 \bar{Q}_1 + Q_2 \bar{Q}_0$$

$$Q_3^* = Q_3 \bar{Q}_1 + Q_2 Q_1 Q_0$$





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